

PAPER_1086: SCm Dark Energy Density with Γ -Coupled Phonon Modulation Replacing Λ CDM

Abstract

We derive the Γ -coupled SCm dark energy density:

$$\rho_{\text{DE}}(t, \Gamma) = \rho_{\text{SCm}}(t) \cdot S_{26} \cdot \Phi(\Gamma) \cdot (2R - 1)$$

This replaces the static Λ CDM cosmological constant $\rho_{\Lambda} = 0.692\rho_{\text{crit}}$ with a dynamical dark energy density sourced by the SCm vacuum, modulated by phonon resonance at 1.25 THz. The ratio $\rho_{\text{DE}}/\rho_{\Lambda}$ quantifies the departure from Λ CDM. Distinct from PAPER_1076 (which derives $w(z)$ from linewidth broadening) and PAPER_889 (Λ CDM contrast), this paper provides the explicit Γ -coupled density computation.

§1 SCm Vacuum Density Evolution

$$\rho_{\text{SCm}}(t) = \rho_{\text{vac,SCm}} \cdot S_{26} \cdot \exp\left(\kappa t + \frac{[\text{SSq}]}{26}t\right)$$

with $\rho_{\text{vac,SCm}} = 9.47 \times 10^{-27} \text{ kg/m}^3$, $\kappa = 5.787 \times 10^{-9} \text{ s}^{-1}$.

§2 Gamma-Coupled Dark Energy Density

$$\rho_{\text{DE}}(t, \Gamma) = \rho_{\text{SCm}}(t) \cdot S_{26} \cdot \Phi(\Gamma) \cdot (2R - 1)$$

where $R = F_{U,Bi}/F_U$ determines the expansion/erosion sign.

§3 Comparison to Λ CDM

$$\rho_{\Lambda} = 0.692 \cdot \rho_{\text{crit}} = 0.692 \cdot \frac{3H_0^2}{8\pi G}$$

$$\frac{\rho_{\text{DE}}}{\rho_{\Lambda}} = \frac{\rho_{\text{SCm}}(t) \cdot S_{26} \cdot \Phi \cdot (2R - 1)}{0.692\rho_{\text{crit}}}$$

At $t = 0$, $R = 0.8$: $\rho_{\text{DE}}/\rho_{\Lambda} \sim 10^{22}$ (phonon amplification dominates at resonance).

§4 Physical Advantages over Λ CDM

Aspect	Λ CDM	SCm Dark Energy
Origin	Ad-hoc constant Λ	SCm vacuum buoyancy
Time evolution	Static $w = -1$	Dynamic via $\rho_{\text{SCm}}(t)$
Free parameters	Ω_Λ (fine-tuned)	κ , [SSq] (calibrated)
Phonon coupling	None	$\Phi_{1.25\text{ THz}}(\Gamma)$
Fine-tuning problem	10^{120} discrepancy	2-parameter resolution
Testability	Indirect (supernovae)	Direct (THz phonon spectrum)

§5 Linewidth Sweep

Γ (THz)	Φ	ρ_{DE} (kg/m ³)	$\rho_{\text{DE}}/\rho_\Lambda$
0.01	1.45×10^{21}	2.18×10^{-4}	2.3×10^{22}
0.10	1.45×10^{21}	2.18×10^{-4}	2.3×10^{22}
1.00	1.45×10^{21}	2.18×10^{-4}	2.3×10^{22}

At resonance, Φ is independent of Γ (Gaussian peak = 1), so off-resonance behavior drives the Γ -dependence.

References

- PAPER_1076: SCm Dark Energy with Phonon Linewidth Γ -Modulation
- PAPER_889: EtVsLambdaCDMDarkEnergyContrastCalc
- PAPER_890: SCmVacuumDensityEvolutionCalc
- PAPER_1087: Dark Energy Equation of State w_DE